

Problem #3

Numerical Results:

```
init=[10^(-3); 10^(-3); 10^(-3); 10^(-3); 10^(-3)];
func = @nlfunc;
jf=@jacob;
format long
xanal=nr_analyticalj(func,jf,init);
fprintf(['Numerical Results: \n' ...
    '[CO2] = %.6f \n' ...
    '[HC03] = %.6f \n' ...
    '[CO3] = %.10f \n' ...
    '[H] = %.10f \n' ...
    '[OH] = %.10f \n'],xanal(1),xanal(2),xanal(3),xanal(4),xanal(5))
```

```
Numerical Results:
[CO2] = 0.001000
[HC03] = 0.001999
[CO3] = 0.0000003997
[H] = 0.0000002507
[OH] = 0.000000399
```

```
options = optimoptions('fsolve','Algorithm','levenberg-marquardt', ...
    'OptimalityTolerance',1.0e-12,'MaxFunctionEvaluations',1000000);
xfsolve=fsolve(func,init,options);
```

Equation solved.

fsolve completed because the vector of function values is near zero as measured by the value of the function tolerance, and the problem appears regular as measured by the gradient.

<stopping criteria details>

```
fprintf(['fsolve results: \n' ...
    '[CO2] = %.6f \n' ...
    '[HC03] = %.6f \n' ...
    '[CO3] = %.10f \n' ...
    '[H] = %.10f \n' ...
    '[OH] = %.10f \n'],xfsolve(1),xfsolve(2),xfsolve(3),xfsolve(4),xfsolve(5))
```

```
fsolve results:
[CO2] = 0.001168
[HC03] = 0.001831
[CO3] = 0.0000002876
[H] = 0.0000003197
[OH] = 0.0001682527
```

Conclusion:

```
percenterror=zeros(5,1);
for i=1:5
    percenterror(i) = 100*abs((xanal(i)-xfsolve(i))/xfsolve(i));
end
```

```
fprintf(['Percent error using Newton-Rhapson results as experimental value,' ...
' and fsolve results as theoretical value: \n' ...
'Percent error for [CO2] = %.2f \n' ...
'Percent error for [HCO3] = %.2f \n' ...
'Percent error for [CO3] = %.2f \n' ...
'Percent error for [H] = %.2f \n' ...
'Percent error for [OH] = %.2f \n'],percenterror(1),percenterror(2),percenterror(3),percenterror(4),percenterror(5));
```

```
Percent error using Newton-Rhapson results as experimental value, and fsolve results as theoretical value:
Percent error for [CO2] = 14.38
Percent error for [HCO3] = 9.17
Percent error for [CO3] = 38.95
Percent error for [H] = 21.57
Percent error for [OH] = 99.98
```

Solution pH:

```
pH=-log(xfsolve(4))
```

```
pH =
14.955935478140626
```

```
function [f] =nlfunc(x)
CO2=x(1);
HCO3=x(2);
CO3=x(3);
H=x(4);
OH=x(5);

f=zeros(5,1);
f(1) = (H*HCO3/CO2)-10^(-6.3); %k1 function
f(2) = (H*CO3/HCO3)-10^(-10.3);%k2 function
f(3) = H * OH - 10^(-14);%kw function
f(4) = CO2 + HCO3 + CO3 - 3*10^(-3);%Ct function
f(5) = HCO3 + 2*CO3 + OH - H - 2*10^(-3);%Alk function
end

function [x] = nr_analyticalj(f,j,x0) % function, jacobian, and initial point
x=x0;
i=0; maxitr=1000000; epsilon=inf;
while (i < maxitr && epsilon > 1e-20)
    i=i+1;
    J=j(x);
    b=f(x);
    dx=-J\b;
    x=x+dx;
    epsilon=norm(dx);
end

end

function [J] = jacob(X)
J = zeros(5);
```

```

x=XX(1);%C02
y=XX(2);%HC03
z=XX(3);%C03
i=XX(4);%H
h=XX(5);%OH

%Jacobian:
J(1,1)=-i*y/(x^2);J(1,2)=i/x;J(1,4)=y/x;%k1

J(2,2)=-i*z/(y^2);J(2,3)=i/y;J(2,4)=z/y;%k2

J(3,4)=h;J(3,5)=i; %kw

J(4,1)=1;J(4,2)=1;J(4,3)=1;%Ct

J(5,2)=1;J(5,3)=2;J(5,4)=-1;J(5,5)=1;
end

```